

Background of Faculty Activity Monitoring and Space Utilisation Analysis

The way of tracking the faculty activities and analysing space utilisation has dramatically changed over the years. This section gives a comparison between the old-fashioned approach and the newly developed IoT and cloud-based approach and the benefits and limitation associated with both.

Traditional Approach:

The process of monitoring the use of the classroom and faculty activities was traditionally conducted manually. Administrative workers had to visit the classrooms physically in order to check the attendance and trace utilisation by using paper-based schedules or spreadsheets. Reports were usually done on a periodic basis, and this implied that the information was retroactive and not real time. Although this was an inexpensive and simple method to apply, it was time consuming, labour intensive and subject to human error. Also, it did not give a lot of insight on the trends of utilisation of space and hence the management could hardly optimise their classroom allocation and future demands. Consequently, overcrowded classrooms, inadequately used facilities, and poor use of energy were a few of the problems that were often detected only when they got out of control.

Modern IoT and Cloud-Based Approach:

Modern solutions, on the contrary, use IoT sensors, cloud computing, and data analytics to deliver a more efficient and automated process. Infrared detectors are the sensors that continuously record the information on occupancy and send it to cloud services in real-time to be processed and stored. The data can be checked by means of the services such as AWS IoT Core, Lambda, DynamoDB, and SNS and authenticated, stored and analysed on the spot, which means that administrators can be notified of the violation of the threshold in a fully automated manner. In addition, the cloud platforms also support sophisticated analysis through the combination of services like Glue and Athena, where long-term trend analysis can be done by querying historical data. This solution removes the necessity of manual surveillance, decreases the administrative load, and improves accuracy. More to the point, the system facilitates the proactive decision-making as it will provide real-time information regarding the classroom

usage, smartly space utilisation, enhancing energy conservation and enabling the overall improvement of the learning environment.

In general, the traditional approach was quite simple; however, it was not efficient and reactive. The current IoT and cloud-based solution, in turn, brings the automation, scalability, and intelligence, so the faculty management could make data-driven decision-making in time.

History of Faculty Activity Monitoring and Space Utilisation Analysis

The history of faculty activity monitoring and space utilisation analysis can be traced back to the early days of university management. During that time, all related processes were performed manually and relied heavily on human observation. The staff responsible for these tasks were usually members of the university's administrative team. They had to inspect classrooms and facilities one by one to ensure they were being properly utilised. If a classroom became overcrowded or if issues such as faulty lighting or air conditioning occurred, the lecturers or students had to report them before the staff could take any action. After study hours, the staff also needed to physically check every classroom and facility to ensure that electricity, lights, and air conditioning were switched off. At the same time, they depended on paper-based records, such as classroom timetables, to determine which courses were using specific facilities. However, these conventional methods cause to several issues. The processes were time-consuming, lacked real-time insights, and were prone to human error, as staff could easily miss or overlook certain details. Furthermore, since these responsibilities were not always handled by the same person, there were inconsistencies standard in defining the utilisation of classroom and facilities.

With the advancement of technology, universities gradually shifted their operations from manual and paper-based approaches to digital solutions. These changes began with simple tools such as spreadsheets and basic database systems, which allowed records to be stored in a more structured way. By integrating IoT sensors and data analytics, the systems significantly improved by support real-time monitoring of space usage within faculty buildings. Since the classroom and facility maintenance is one of the highest expenditures for universities after lecturers' salaries, the adoption of space utilisation analysis not only helps reduce operational costs but also provides valuable insights for strategic management. As a result, it improves the student experience by preventing overcrowding while ensuring greater efficiency in the management of resources and facilities.

In conclusion, the advancement of faculty activity monitoring and space utilisation analysis plays a crucial role in helping universities to optimise their resources and support sustainable institutional growth.

Future Trend of Faculty Activity Monitoring and Space Utilisation Analysis

As time goes on, higher education continues to evolve in order to meet institutional demands and adapt to the changing needs of students. Therefore, there are several key trends that are expected to shape the future of Faculty Activity Monitoring and Space Utilisation Analysis. Among these, the development of artificial intelligence (AI) and predictive analytics, along with the advancement of Internet of Things (IoT) sensor technology are becoming more sophisticated and comprehensive. These technologies have the potential to maximise efficiency and reduce costs, thereby contributing to improved faculty performance and enhancing the overall student experience.

One of the key trends is the development of smart campuses through IoT integration. For instance, smart lighting and air-conditioning systems equipped with IoT sensors can automatically detect occupancy in classrooms and facilities. When a room is occupied, the systems are activated, and while it is empty, they switch off. As a result, less energy is used, which lowers operating expenses. Furthermore, this trend also introduces the concept of the “smart classroom” in higher education, where resources can be automatically adjusted based on the number of students present. Thus, this fosters a more efficient and responsive learning environment.

In addition, the development of artificial intelligence (AI) and predictive analytics is also one of the key trends. These technologies can forecast and predict peak classroom demand, as well as help balance teaching schedules and room allocations. With their assistance, the administrative burden on the staff can be reduced while simultaneously improving the overall efficiency. Furthermore, AI can support decision-making by providing the staff with real-time insights into facility usage and resource availability. As a result, institutions are able to adopt more data-driven planning and management strategies that can lead to better resource optimisation.

Problem Statement

Educational institutions face inefficiencies in the management of their physical classroom spaces. The core issue lies in the reliance on outdated, manual processes for monitoring and controlling classroom environments. Staffs are required to physically patrol campus buildings to check on occupancy, adjust lighting, and regulate heating, ventilation, and air conditioning (HVAC) systems. This labour-intensive and time-consuming approach wastes time and resources. Without real-time, granular data on how spaces are being used, resources are frequently wasted, and learning environments are not optimised to meet immediate needs.

The consequences of the manual, inefficient system are multi-faceted and directly affect financial, operational, and environmental aspects of the institution:

1. Financial and energy waste
 - Unoccupied or underutilised classrooms often remain fully lit or climatized. This leads to substantial and avoidable energy consumption, contributing to inflated utility bills and a larger carbon footprint.
2. Suboptimal learning environments
 - Manual control systems are slow to respond to changes in classroom conditions. As a result, students and instructors may experience uncomfortable temperatures, poor lighting, or stuffy air, which can negatively impact concentration, engagement, and overall academic performance.
3. Slow strategic planning
 - Without accurate, historical data on classroom usage, facility managers and administrators lack the insights needed for effective long-term planning. Decisions regarding new construction, renovation, or the decommissioning of buildings and classrooms are made based on anecdotal evidence rather than actual usage patterns, leading to misallocated funds and underutilized assets.
4. Security and safety concerns
 - A lack of real-time occupancy data can also present safety and security challenges, as there is no automated way to detect or respond to potential overcrowding situations or to track who is in a building after hours.

Objective and Proposed Solution

- Develop a tool that makes use of AWS that can help the university manage their resources.

The Amazon Web Service (AWS) cloud provider will be used to develop the tools for monitor faculty's activity. This tool aims to facilitate the management's goal to optimise space utilisation – some areas are overcrowded while others remain consistently empty. By providing real-time and historical data on occupancy, activity levels, and peak usage times, the system allows faculty administration to identify underutilized classrooms, labs, offices, and study areas. This data can then inform decisions on re-purposing spaces, optimizing timetabling, or justifying investment in new facilities.

The system would also allow the faculty's administration to make better informed decision on which facility to invest. This allows them to manage their finances much better – choose which areas or classrooms to renovate. For instance, cleaning schedules can be optimized based on actual traffic, IT resources can be deployed to high-usage areas, and future campus development can be tailored to demonstrated needs.

Finally, the system would provide real-time or manual adjustments of facilities. For example, turning off the air-conditioner, lights, or fans when no one is using classrooms or meeting rooms.

Architecture Diagram of Proposed Solution

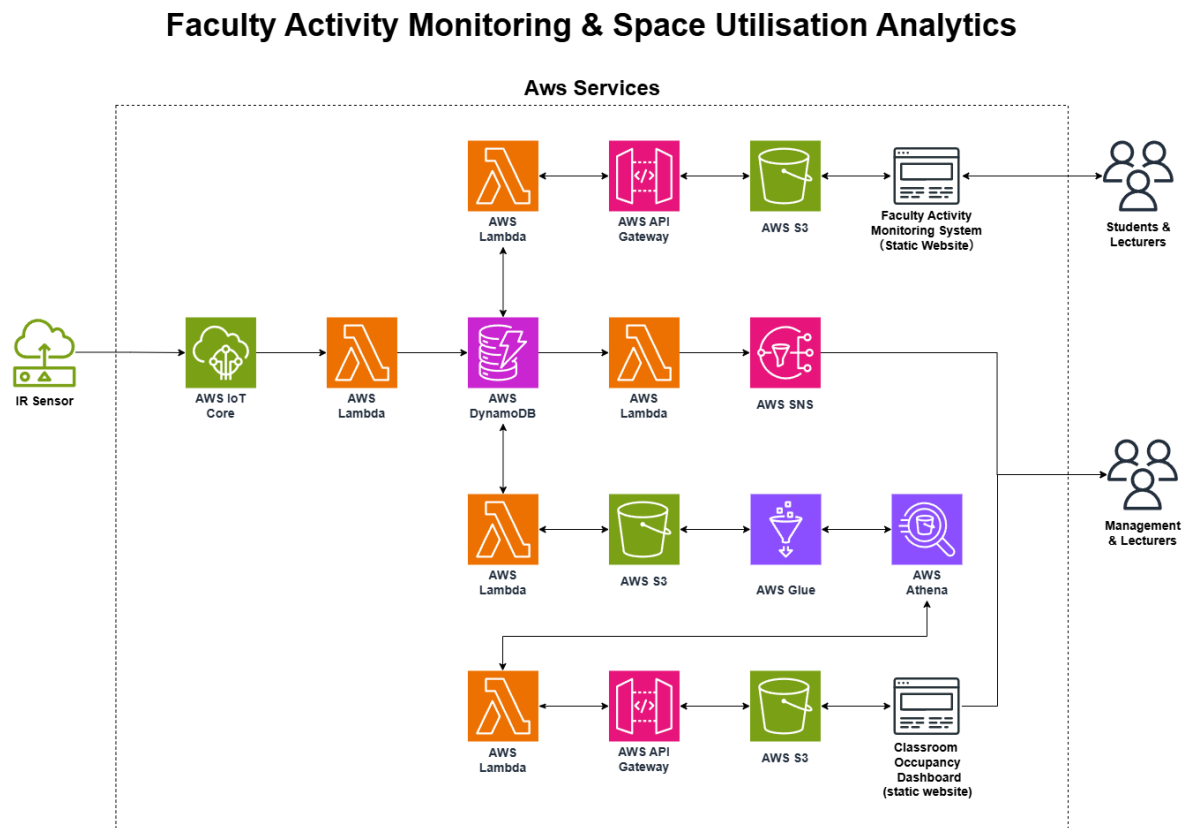


Figure 1.1: Architecture Diagram of Proposed Solution

Figure 1.1 shows the proposed architecture for the “Faculty Activity Monitoring and Space Utilisation Analysis” system. The design comprises eight distinct component types deployed on the Amazon Web Services (AWS) platform, and the IR sensor was simulated using Node-RED, and the end users are students, lecturers, and management. The following section discusses each component in detail, and the section concludes with an overview of the overall design flow.

Firstly, **Amazon IoT Core** is one of the components offered by Amazon that captures input from sensors. Through this cloud service, users can connect their input sensors to a variety of Amazon applications. The cloud service introduces many different communication protocols to allow users to establish a secure connection, including MQTT, HTTP, and WebSockets. Even more, IoT Core offers various authentication mechanisms, including using tokens, certificates, or IAM policies, to guarantee the security of the connection. When IoT Core is receiving incoming data from a sensor, it will perform the authentication using a method predefined by users. Once authentication is complete, IoT Core directs the data to other Amazon services that

users have specified. For instance, the IoT Core in our solution uses MQTT and WebSocket to establish a connection with the sensors to receive the data while using certificates and IAM policies to verify the sensors.

In addition, **Amazon Lambda** is the most widely used component in this system to perform specific functionalities. Users are allowed to customise their functionality by deploying their code directly in Lambda. As a serverless compute service, Lambda eliminates the need for users to manage the servers. All configured functions can automatically run the code without any human intervention. To achieve this, users are required to configure an event trigger which calls the function when a predefined event occurs. For example, some of the Lambda functions in this design respond to events generated by DynamoDB. When there is any update on the specified table, the corresponding Lambda function automatically begins processing the request. More importantly, the automatic scaling feature that Lambda offers allows it to launch more function instances during high demand and reduce them when idle and pay-per-execution allows it to optimise its costs.

Moreover, **Amazon S3** in this architecture is mainly for storing the assets of static websites. It is an object storage service designed to store and retrieve raw data and commonly applied as a data lake, backup, and file storage. To store objects into S3, users must create a bucket in S3 to hold users' data as a storage container. Users and applications can then upload and manage their S3 objects using the AWS Console, API, or SDKs. Besides, S3 can also be utilised as a static website hosting service. As long as users enable public access, other users will be able to retrieve their uploaded assets, such as HTML, CSS, JavaScript, and media files, directly from the S3 bucket. As an example, two static websites have been hosted for this solution, both utilizing API Gateway to retrieve data from DynamoDB. The utilization of S3 is more cost-effective in hosting static websites compared to EC2. It is because users are only paying for the storage and data transfer and not for the computing cost while the website is sitting idle.

Apart from that, this system uses **Amazon SNS** to alert administrators when the classroom reaches its maximum capacity. SNS is a publish/subscribe messaging service that allows the subscribers to receive messages sent by the publisher. To receive messages, users can subscribe to the SNS topics and select the methods, such as mobile push, SMS, and email, to receive them. In detail, publishers can be any application, service, or AWS component, while a topic serves as the middleware that connects them to subscribers in SNS. The communication processing starts when the publisher produces the message and sends it to the SNS topic. The

SNS topic then sends it to all the subscribers that are subscribed to the topic. In this case When the classroom is full, the publisher (Lambda) will send a message to the SNS topic. The SNS service will then send that message to all subscribers according to their delivery methods.

On top of that, **Amazon DynamoDB** is one of the database services provided by Amazon. Different from traditional relational databases (Amazon RDS), DynamoDB uses key-value pairs or documents (JSON) to store data. In RDS, a fixed schema model is used, where tables, rows, and columns must be predefined. In contrast, DynamoDB allows flexible storage of attributes because it uses a schema-less model, allowing users to add new attributes without affecting the table structure in the future. DynamoDB is more suitable in this case because large amounts of throughput can be processed with lower latency while allowing real-time event-driven processing, which is a better option for storing large amounts of sensor data. To enable real-time event-driven processing, DynamoDB Stream must be enabled. It is mainly used to capture the changes of the table and trigger other services in real time. To illustrate, when the “Classroom” table is modified, a Lambda function is triggered to check the current capacity. If count exceed the classroom limit, notification is sent to the admin using SNS.

Additionally, **API Gateway** is essential for managing and routing API requests. It is also in charge of ensuring that all client access to backend services is secure. Numerous procedures for developing, publishing, sustaining, monitoring, and safeguarding APIs are offered by API Gateway. In order to authenticate and validate the request, the client will first send it to the API Gateway. The request is routed to the backend services for processing after authentication. Among the backend services are EC2, S3, and Lambda. The client will receive the response in the format of their choice after the backend service responds. To satisfy user needs, a variety of API types, including REST, HTTP, and WebSocket, are also supported. For instance, the hosted static website retrieves data using a REST API. The hosted static website simply needs to retrieve and show data, therefore even though REST APIs allow GET, POST, PUT, and DELETE methods, only the GET method is employed.

Furthermore, **Amazon Glue** is used in this system to simplify the process of preparing the data. It is an ETL (Extract, Transform, Load) service mainly for moving and transforming data. It allows applications to retrieve various sources of data from databases, data lakes, and application logs. The process begins with data discovery, where sources are scanned by Glue crawlers and schema information is stored in the Glue Data Catalog. After that, users can create ETL jobs to process the data. The transformed data is stored in destinations and ready for

querying. Last but not least, **Amazon Athena** is added into this system to query and analyse data stored in Amazon S3. It is responsible for making massive datasets to be accessed securely. While allowing users or applications to use SQL to interact with it. Athena also offers many options for submitting SQL queries, including APIs, Athena console, and drivers. After submission, Athena employs a distributed engine to authenticate the user while managing the submitted requests and validating the queries. The results are returned in the user-defined format after the data in S3 has been received and filtered. Additionally, Athena can also work with Amazon Glue. With this, Athena can automatically retrieve schema and metadata from the Glue Data Catalog; therefore, queries are not only more precise but also more manageable. In particular, a hosted monitoring dashboard in the system uses Athena to query the occupancy and classroom details, while Glue manages the schema automatically.

The following explains the usage and flow of each of the above services in our solution. Firstly, the IoT Core is used to collect data from IR sensors. It uses MQTT and WebSocket to establish a connection with the sensors to receive the data while using certificates and IAM policies to verify the sensors. Once authenticated, the incoming data is processed using Lambda and stored in DynamoDB. The rest process can be divided into three parts, from top to bottom. The first part is to allow users (students and lecturers) to check the current capacity of each classroom, as shown in figure 1.2. When the users click on the static website (Faculty Activity Monitoring System) that is hosted on S3, the website sends an HTTP GET request to API Gateway. Next, the API Gateway requests the Lambda to fetch the data from DynamoDB. The data is then formatted and passed back to the API Gateway in JSON file format. Once the website receives the data, it processes and displays the data on the client side using JavaScript. Meanwhile, the second part is used to alert management and lecturers when the occupancies reach the limit. The Lambda function is predefined with a trigger, and when the table in DynamoDB is updated, the trigger is fired and starts evaluating the occupancies. If the count exceeds the limit, SNS is triggered to send a notification to management and lecturers. Lastly, the third part is designed to monitor and analyse the occupancy over time and classroom status (Classroom Occupancy Dashboard), as illustrated in figure 1.3. Similar to the second part, the Lambda connected to the DynamoDB is triggered upon any updates to the table and subsequently updates the data in S3 each time it is invoked. While the website part is also similar to the first part, the only difference is that the Lambda gets the data from S3 using Athena and Glue instead of directly accessing DynamoDB.

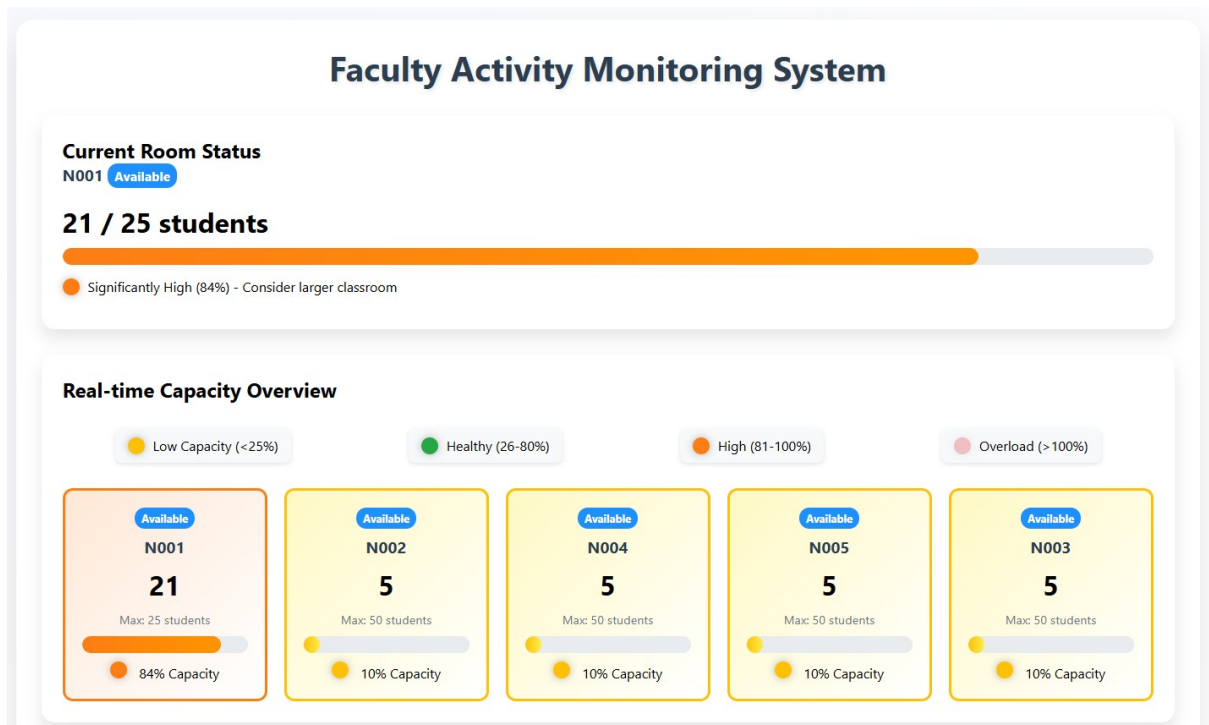


Figure 1.2: Faculty Activity Monitoring System



Figure 1.3: Classroom Occupancy General Dashboard

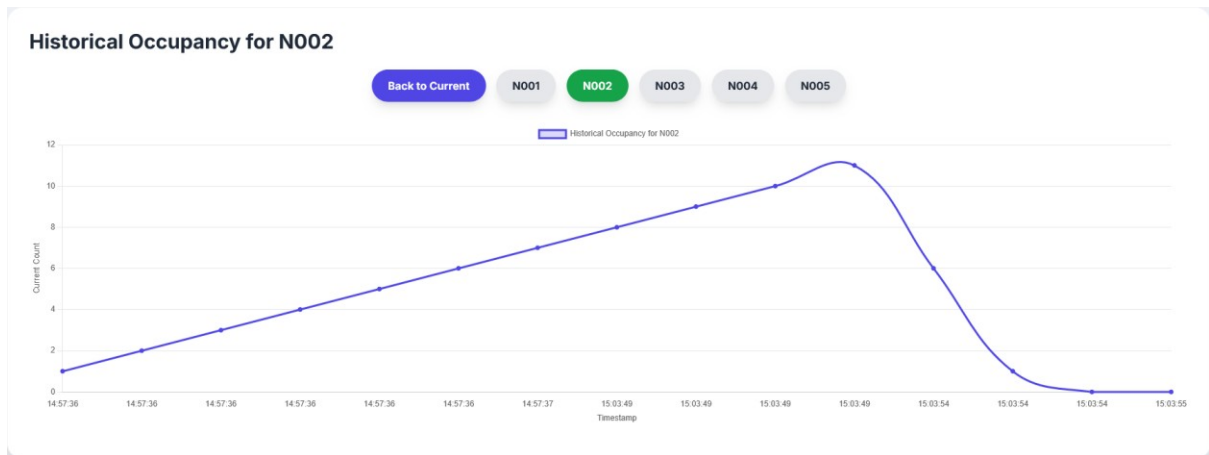


Figure 1.4 Class Historical Occupancy

Classroom Status Overview

CLASS ID	COUNT	MAX CAPACITY	STATUS	ALERT LIGHT	SUGGESTION
N001	27	25	OVERLOAD (108%): Capacity exceeded!	●	ALERT MODERATOR! Suggest another classroom with suitable capacity.
N002	74	50	OVERLOAD (148%): Capacity exceeded!	●	ALERT MODERATOR! Suggest another classroom with suitable capacity.
N003	54	50	OVERLOAD (108%): Capacity exceeded!	●	ALERT MODERATOR! Suggest another classroom with suitable capacity.
N004	6	50	Low Level (12%): Nearly empty.	●	Suggest a suitable, smaller class to be used.
N005	2	50	Low Level (4%): Nearly empty.	●	Suggest a suitable, smaller class to be used.

Figure 1.4 Class Status Overview

Study of Different IoT Platforms

Google IoT

Google Cloud is a cloud computing service provider operated by Google. Similar to AWS, Google Cloud follows a pay-as-you-go pricing model. Google Cloud offers over 150 products and services across categories such as artificial intelligence (AI) and machine learning (ML), data analysis, website development, application development, and industry-specific solutions. The featured products by Google Cloud are Vertex AI platform to train AI and ML applications; Compute Engine for virtual machines (VMs); BigQuery for data warehouses; Cloud Run for serverless computation; Cloud SQL to host SQL databases; and Cloud Storage for object storage.

Cloud IoT Core is the service that is responsible for managing IoT devices. With this service, users can register, authenticate, and authorise IoT devices and users inside the Google Cloud resource hierarchy. Additionally, with this service, IoT devices' metadata can be stored in the cloud, and practitioners are given the ability to **send device configurations** to other **Google Cloud Platforms (GCP) or third-party services to devices**. Cloud IoT Core supports **standard protocol bridges – MQTT and HTTP – for bi-directional communications, automatic load balancing**, and **global data access** with Cloud Pub/Sub service – messaging backbone for IoT data. Moreover, Cloud IoT Core is equipped with industry-standard security protocols, ensuring strong security.

This service can be used to **trigger Cloud Functions** by forwarding device telemetry data to **Cloud Pub/Sub service**. Cloud IoT Core also **supports streaming analysis**, whereby users can use **Cloud Dataflow** – real-time data analysis platform, **similar to AWS Glue or AWS Data Pipeline** – or custom analysis with custom subscribers. Both direct connections and gateway-based architectures are fully supported by Cloud IoT Core.

Furthermore, **Cloud IoT Edge**, a companion to Cloud IoT Core within the Google Cloud IoT platform, enables **local data processing and intelligence at the edge**. With Cloud IoT edge, users can utilise Google Cloud's AI capability to gateways and connected devices, allowing the development of ML models on the cloud and be executed on Cloud IoT Edge devices.

AWS IoT

AWS (Amazon Web Services) is a well-known cloud computing platform provided by Amazon. It provides over 200 services that allow users to run applications or store data without the need for physical infrastructure. At the same time, users only pay for the resources they use based on demand. Within this IoT ecosystem, AWS IoT Core plays an essential role since it offers a managed cloud platform that enables secure and two-way communication between IoT devices and the cloud. This service simplifies the process of collecting, processing, and modifying on data from connected devices. By integrating with other services such as Amazon Kinesis for real-time data streaming, AWS Lambda to handle serverless event-driven processing, and so on, users can build a robust IoT solution without managing backend infrastructures.

Besides AWS IoT Core, AWS provides a few specialized services that combine to offer a complete solution for various use cases. The first category is **Connectivity and Control Services**, which forms the foundation of AWS IoT solutions. For example, **IoT Core** guarantees secure messaging, **IoT Device Management** allows organizations to register, organize, and remotely update devices at scale, **IoT Device Defender** performs continuous audits and anomaly detection to strengthen security, while **Amazon Kinesis Video Streams** supports real-time video streaming from connected cameras or sensors.

For the second category, it is called **Edge Services**. These services ensure applications continue to function even with poor internet connectivity by directly extending AWS capabilities to devices. For examples, **FreeRTOS** allows microcontroller-based devices to connect safely to the cloud, **AWS IoT Greengrass** which permits local data processing and machine learning inference on edge devices, and **IoT ExpressLink** that makes cloud connectivity easier through prebuilt secure hardware modules.

To address industry-specific requirements, AWS introduces **specialized IoT services** to tailor some specific domain. **AWS IoT SiteWise** helps to collect and analyse data from industrial equipment to improve operational efficiency. **AWS IoT TwinMaker** supports the creation of digital twins for real-world systems like factories and buildings to enable visualization and simulation. **AWS IoT FleetWise** is designed for collecting and analysing vehicle data in the automotive sector.

In summary, the above services make AWS become one of the most complete and scalable IoT platforms.

Azure IoT

Azure IoT is a cloud computing service that enables Microsoft to offer a fully-fledged infrastructure to connect, monitor and manage large-scale Internet of Things (IoT) devices. It combines software, hardware, and cloud services to facilitate a fluent interaction between devices and the cloud which allows real-time data gathering, storage, analysis and decision making. The ability to remotely operate devices with simultaneous valuable insights about generated data is provided by Azure IoT, which offers organizations secure and bi-directional communication and empowers them to take control of devices remotely.

The **main role of Azure IoT** is to convert raw sensor data into an actionable intelligence. The platform supports device connectivity, predictive analytics, process automation, and the development of a physical environment replica through its basic services, including IoT Hub, IoT Central, and Azure Digital Twins. Through these capabilities, organizations will be able to increase operational efficiency, better decisions, and create new data-driven business models in line with the current digital transformation strategies.

Azure IoT has found its applicability in **numerous critical industries** which include manufacturing, healthcare, energy, logistics and smart cities. It can be used to predictive maintain and production optimization in manufacturing, remote patient monitoring and medical equipment integration in healthcare, grid optimization and integration of renewable energy sources in energy, asset tracking and supply chain visibility in logistics, and intelligent infrastructure, including traffic control and environmental surveillance and citizen safety, in smart cities.

Key Features of Azure IOT

Azure IoT Hub is a cloud service that allows bidirectional communication between millions of IoT devices and cloud-based solutions. It allows developers to monitor and develop custom solutions. In addition, IoT Hub can integrates seamlessly with other Azure services including Azure Stream Analytics, Azure Functions, and Azure Machine Learning, allow advance processing, analytics, and AI based insights across IoT deployments. Azure IoT Hub also offers programmable SDKs for popular languages such as Java, Python, and Node.js and supports

multiple protocols, including MQTT, AMQP, and HTTP/HTTPS to meet different requirements from developers.

Azure IoT Central is a fully managed IoT application platform that functions as a software-as-a-service solution (SaaS). It simplifies the development process of IoT applications by providing industry-specific application templates to users. After successfully building the solution, users can execute use cases such as connecting, managing, and monitoring IoT devices. IoT Central is suitable for users who prefer not to develop applications from scratch because they are not required to have deep technical knowledge about IoT solutions. Interestingly, this platform builds on top of over 30 other services, using IoT Hub as the fundamental stepping stone.

Azure IoT Edge is a service that deploys the cloud intelligence on IoT edge devices. It allows cloud intelligence such as Azure services, custom logic, and artificial intelligence (AI) to run locally on connected devices. With this, the devices can operate and interact securely even without an internet connection. This approach is also known as “edge computing”, which also helps in reducing latency and conserving bandwidth. In addition, Azure IoT Edge offers an AI toolkit that simplifies the execution of machine learning and AI capabilities. For example, image recognition and complex processing, without the need to write complex code.

Azure IoT Solution Accelerators is a solution that help developers to quickly build their IoT solutions. It offers pre-build, ready-to-deploy templates to accelerate the development. For instance, it comes with pre-built dashboards for data visualization, built-in device connectivity for secure communication, and seamless integrations with other Azure cloud services for data processing and storage. Moreover, IoT Solution Accelerators also provides customisable foundation for specific industry or business needs, eliminating the need to build from scratch.

Comparison of Relevant Technologies

The analysis of various IoT solutions demonstrates the advantages and drawbacks of Google IoT, AWS IoT, and Azure IoT in terms of supporting real-time monitoring of faculty activity and space utilisation analysis. Although all three of these platforms can handle the IoT devices and allow them to operate in the cloud-based analytics, they are not equally suitable in terms of scalability, integration, and long-term sustainability.

Google Cloud IoT Core used to provide a stable device management and secure connectivity as well as connections to other services like BigQuery and Dataflow to provide analytics. Its advantage was that Google ecosystem of artificial intelligence and machine learning products, specifically Vertex AI and TensorFlow. Nevertheless, the fact that Cloud IoT Core was discontinued in August 2023 greatly reduces its applicability as a long-term solution to IoT deployments. Google Cloud is still competitive in analytics but is less competitive in solutions with a need to monitor devices continuously and at scale since it does not have a dedicated IoT device management service.

AWS IoT offers the most tuned and full service of IoT. As the key central service, IoT Core will allow two-way communication between the devices and the cloud to occur in a secure way. It can be used in conjunction with other AWS services like Lambda, DynamoDB, SNS, Glue and Athena which are already implemented in this project. This renders AWS very appropriate to event-driven frameworks in which real-time notifications and automated processing is essential. AWS also has a wide range of edge computing global support with IoT Greengrass and FreeRTOS, and industry-specific applications (SiteWise, TwinMaker and FleetWise). It has a mature, reliable, and popular ecosystem in the academic and industrial spheres; thus, it is the best candidate to use in this project.

Azure IoT can be described as very competitive, and it provides a wide range of tools, such as IoT Hub, IoT Central, and Digital Twins. It is powerful in terms of enterprise adoption in such industries like healthcare, energy, logistics, and smart cities. Depending features like IoT Edge are enabling advanced AI and ML processing directly on devices and therefore reducing latency and relying on cloud connectivity. In addition, Azure IoT Central eases the process of deploying an IoT solution providing SaaS templates. Solutions provided by Azure might however be more complex to set up and integrate, which is not necessarily ideal in small-scale or academic builds.

Overall, AWS IoT is best suited to operate in the Faculty Activity Monitoring and Space Utilisation Analysis given its scalability, real-time event management and smoothness with analytics and alerting services. Google IoT, which was powerful in analytics, has lost its viability because of the closure of the IoT Core. Azure IoT offers robust enterprise functionality, yet it is complex and expensive and will be of little use in this academic setting. Table below shows the cross-comparison of the feature of each IoT:

Table 1.0: Cross-comparison of the feature of each IoT

Feature / Aspect	Google IoT	AWS IoT	Azure IoT
Core Service	Cloud IoT Core (device management & telemetry, retired in Aug 2023)	AWS IoT Core (secure two-way communication, scalable)	Azure IoT Hub (bi-directional communication for millions of devices)
Edge Capabilities	Cloud IoT Edge (runs ML at edge using Google AI models)	AWS IoT Greengrass, FreeRTOS (local processing, ML inference on edge devices)	Azure IoT Edge (deploys AI, analytics, and services to devices without internet)
Data Processing & Analytics	BigQuery, Dataflow , Pub/Sub (streaming, warehousing, analytics)	Kinesis, Glue, Athena (streaming, ETL, serverless SQL)	Azure Analytics, Learning, Twins (real-time analysis & simulation)
AI/ML Integration	Strong (Vertex AI, TensorFlow integration)	Moderate (SageMaker for ML models, Greengrass ML inference)	Strong (Azure ML + AI Toolkit for edge)
Device Management	Supports registration, authentication, metadata storage	IoT Device Management (register, organize, update devices at scale)	IoT Central (SaaS model for easy deployment, templates)
Industry-Specific Solutions	General purpose, strong in data/AI workloads	Specialized services (IoT SiteWise, TwinMaker, FleetWise)	Widely adopted in healthcare, energy, logistics, smart cities
Security	IAM, certificates, token-based	IoT Device Defender (auditing, anomaly detection), strong IAM integration	Enterprise-grade security, compliance, access policies
Pricing	Pay-as-you-go, but Cloud IoT Core discontinued	Pay-as-you-go, broad service coverage, scalable	Pay-as-you-go, SaaS options reduce setup effort
Strengths	Excellent analytics & AI ecosystem	Most complete & scalable IoT platform, tight AWS integration	Strong enterprise adoption, templates, and edge AI

Weaknesses	IoT Core service deprecated → less suitable long term	Complex ecosystem, learning curve	Can be costly, requires integration across many Azure services
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